

1 **Shifts in metabolic scaling, production, and efficiency across major evolutionary**
2 **transitions of life.**

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19 One-sentence summary: The scaling of metabolic rate, biomass production, and
20 production efficiency shift across the major evolutionary transitions of life.

The diversification of life involved enormous increases in size and complexity. The evolutionary transitions from prokaryotes to unicellular eukaryotes to metazoans were accompanied by major innovations in metabolic design. Here we show that the relationships between metabolism, production, and size have changed across the evolutionary transitions. Metabolic rate scales with body mass superlinearly in prokaryotes, linearly in protists, and sublinearly in metazoans, so Kleiber's $\frac{3}{4}$ power law does not apply universally across organisms as previously thought. The scaling of maximum population growth rate shifts from positive in prokaryotes to negative in protists and metazoans, and the efficiency of production declines across these groups. As organisms increased in size and complexity, they faced and overcame new constraints that limited metabolic rate and the efficiency of reproduction.

The 3.5 billion year history of life on earth is characterized by dramatic increases in the complexity, diversity, and maximum size of living things. Much of this diversity developed by way of a few major transitions in form and function (1). The first organisms were microbes with a relatively simple body plan and metabolic network. The diversity of contemporary organisms ranges from minute, relatively simple unicellular prokaryotes and archaea to giant, complex animals and plants containing multiple differentiated organelles, cells, tissues, and organs.

Two of the largest transitions in the history of life were from simple prokaryotic to complex eukaryotic cells, and from unicellular to multicellular eukaryotes. Each

transition required the integration of a group of individuals into a new higher-level unit of selection and organization (1). These transitions involved dramatic changes in structure and function, and several orders of magnitude increase in body size (2). Since all organisms share a common set of molecules and biochemical reactions (3, 4), the increases in size and organizational complexity were accomplished by assembling these universal components in new ways (5). Major changes in genetic systems made these transitions possible (1, 6), and complementary changes in metabolic systems supplied the energy and materials to grow larger and support more complex morphologies and physiologies (7, 8).

Scaling relations offer powerful insights into the fundamental processes that constrain and regulate biological structure and function. Nearly all characteristics of organisms, from use of energy to the population growth it fuels, vary with body size. Most of the variation can be described by allometric equations or power functions of the form

$$Y = Y_0 M^\alpha \quad (1)$$

where Y is the trait of interest, Y_0 is a normalization constant, M is body mass in grams, and α is the scaling exponent. There is a large and longstanding literature on these biological scaling relations, although only a few studies of unicellular prokaryotes and protists. Given the influence of the major evolutionary transitions on the biological design of organisms, we hypothesize that the transitions should influence the allometric scaling of three traits that are relevant to the physiology, evolution, and ecology of organisms, which we consider below:

1) Metabolic rate: Metabolic rate, B , the rate of energy transformation, is perhaps the most fundamental biological rate. It sets the pace of life. It is statistically correlated with and functionally linked to many other traits. In the 1930s Max Kleiber (9) showed that the metabolic rate of birds and mammals scales as approximately the $3/4$ power of body mass. Subsequent findings of similar scalings for metabolic rates in many kinds of life forms led to the canonization of “Kleiber’s law”: $\alpha \approx 0.75$ was thought to apply to all organisms, including unicellular prokaryotes and eukaryotes (10-14). Renewed interest in biological scaling relations has led to empirical and theoretical re-evaluation of Kleiber’s law, with much dispute over the exact value of α in different taxonomic and functional groups. Although there is still controversy, α is usually less than 1 in multicellular organisms, signifying that larger multicellular organisms have lower metabolic rates per unit mass. Exceptions to the generality of sublinear scaling are the steeper, near-linear scaling observed for small multicellular organisms (15), small plants (16), and in some studies of prokaryotes and protists (17-19). These results delineate a major empirical and theoretical conflict. Theoretical models have attributed $3/4$ -power scaling to the fractal-like designs of vascular systems of large, complicated organisms (20), but linear scaling is hypothesized to occur because of an optimal metabolic rate for all living things (17). Clearly, the scaling of metabolic rate with body mass in small organisms needs to be re-examined, with a focus on the evolutionary transitions that connects these disparate forms of life.

2) Population growth rate: The rate of population growth, r , is another trait with fundamental importance in both ecology, where it provides a standardized estimate of the

population-level rate of biomass production, and evolution, where it is often taken as a measure of fitness. Population growth rate under optimal conditions, r_{max} , has received considerable attention in both basic and applied studies of microorganisms. Because production of new biomass for both growth and reproduction is fuelled by metabolism, it has generally been assumed that r_{max} scales in the same way as mass-specific metabolic rate, so with an exponent of less than one and approximately -0.25, given that they follow Kleiber's law. This has generally been supported by empirical studies of large, multicellular organisms (12,21). Although a seminal early study of r_{max} in protists reached similar conclusions (22), the scaling of r_{max} across the evolutionary transitions should be re-examined.

3) Efficiency of biomass production: Another basic characteristic of organisms is the efficiency with which they convert metabolic energy into new biomass. This efficiency, E , can be expressed in units of gJ^{-1} as the rate of biomass production divided by the rate of metabolism, both standardized as per unit body mass, so as $E = r_{max} / (B / M)$. E is not only a fundamental biological parameter; it has important practical applications in areas such as agriculture, biotechnology, and biofuel production. So it is timely to quantify the scaling of E as a function of body size and across the evolutionary transitions.

Here we compile data on the scaling of three fundamental characteristics, metabolic rate, B , maximum population growth rate, r_{max} , and efficiency of biomass production, E , in three functional groups of heterotrophic organisms: prokaryotes, protists, and small multicellular aquatic animals (hereafter metazoans). These data accompany this paper, along with details of our criteria for incorporating data (23).

Application of a scaling framework is especially powerful and informative when the organisms vary in body size by many orders of magnitude in body mass. Our data include organisms spanning about 16 orders of magnitude in body size and representing the evolutionary transitions from prokaryotes to unicellular eukaryotes to multicellular animals. To control for the effects of food supply and activity, the metabolic rate data are classified into two categories according to the conditions of under which the measurements were taken: (i) active and fed, and (ii) inactive and without any external source of food, which we refer to as active and endogenous, respectively. These data include 167 and 188 species represented in each state, respectively. We analyze these data in the context of allometric scaling to evaluate our hypothesis that scaling of metabolic rate changed across the evolutionary transitions from small, simple prokaryotes to much larger and more complex metazoans. Using nested ANOVAs, we identify differences in scaling slopes and intercepts among groups. Our findings contradict current dogma about the scaling of metabolism and $r_{\max}$, demonstrate how existing constraints and new innovations affected the evolutionary transitions, and raise exciting new questions about the role of energy in the diversification of life.

Whole-organism metabolic rate increases with body size across prokaryotes, protists, and metazoans, but each group is characterized by a distinctive scaling relationship (Fig. 1). Although the entire dataset can be fit with a single power law that accounts for most of the variation, the relationship between body mass and metabolic rate is significantly improved by incorporating evolutionary group for both active and endogenous rates (ANOVA comparing a 3-line with a 1-line model; active, $F_{4,161} = 9.57$,

$p < 0.0001$; endogenous, $F_{4,182} = 6.07$, $p < 0.0001$). As a check, we also tested for a difference in slopes between protists and metazoans, which differ for both active and endogenous rates (ANOVA comparing a 2-line with a 1-line model; active, $F_{1,119} = 3.87$, $p = 0.05$; endogenous, $F_{1,63} = 3.96$, $p = 0.05$). We used ordinary least squares (OLS) rather than reduced major axis (RMA) estimation because metabolic rate is determined by body mass rather than the other way round, but slope estimates should be corrected upward to allow for error in measuring body mass (24). Figure 1 shows the raw data and the corrected fits and slopes for each group (blue lines and symbols indicate endogenous data, black indicate active data). The corrected scaling exponents for active metabolic rates ($\pm$ SE) are 1.96 ± 0.18 , 1.06 ± 0.07 and 0.79 ± 0.04 for prokaryotes, protists and metazoans respectively. Likewise, for endogenous metabolic rates, they are 1.72 ± 0.07 , 0.97 ± 0.07 and 0.76 ± 0.08 . The distinctive shift in scaling is visible for both active and endogenous metabolic rates, and the slopes for the two physiological states are roughly parallel. This analysis clearly shows that the scaling of metabolic rate with body size changed fundamentally across each of the evolutionary transitions.

The differences across groups and the large discrepancy between the canonical $\alpha = 0.75$ and the observed exponents for protists and especially for prokaryotes clearly show that Kleiber's law, long thought to extend across all living things, does not hold for single-celled organisms. Instead, each of the three groups of heterotrophic organisms exhibits a distinctive scaling of metabolic rate with body mass: superlinear in prokaryotes ($\alpha \approx 1.8$), nearly linear in protists ($\alpha \approx 1$), and sublinear (Kleiber's law) in metazoans ($\alpha \approx 0.75$). These data suggest that scaling of metabolic rate is not governed by a single,

overarching design principle that applies to all living things, but instead by different constraints operating on body sizes and levels of structural and functional organization.

The scaling of $r_{\max}$ also changes across the evolutionary transitions. $r_{\max}$ increases with mass in prokaryotes and then scales negatively in both protists and metazoans (Fig. 2A). This result contradicts previous studies that found $r_{\max}$ scaling with an exponent of approximately -0.25 across diverse taxa from prokaryotes to mammals (22). Mass-specific metabolic rate scales as $M^{\alpha-1}$, so according to the theory that within group $r_{\max}$ is proportional to mass-specific metabolic rate (here considering only active rates), $r_{\max}$ should scale as $M^{-0.8}$ in prokaryotes, M^{-0} in protists, and $M^{-0.25}$ in metazoans. Overall, the scalings of $r_{\max}$ parallel the scalings of mass-specific active metabolic rate as predicted by theory (Fig. 2A, ANOVA comparing 3 parallel-line with 6-line model, $F_{3,331} = 0.13$, *NS*), supporting the idea that within groups biomass production is proportional to mass-specific metabolic rate and is invariant with body mass.

Finally, mean efficiency of converting energy to biomass production decreased with each successive evolutionary transition by approximately an order of magnitude across the three groups, from $23 \times 10^{-4} \text{ gJ}^{-1}$ for prokaryotes to 9.2×10^{-4} for protists, to 1.6×10^{-4} , for metazoans (Fig. 2B, $p < 0.001$). Evidently, the increased whole-organism metabolic rate that accompanies the added levels of organization and larger body size associated with the transitions occurs at the expense of decreased efficiency of conversion of metabolic energy into biomass. The mechanisms underlying this decrease in efficiency with increasing body size and complexity warrant investigation. Larger, more complex organisms allocate relatively more metabolic energy to acquisition and

processing of resources and relatively less to biomass production. We interpret this decrease in efficiency as resulting in part from the transitions of metabolic processes from extra-cellular and surface-localized phenomena in prokaryotes, to organelle-based processes in protists, to complex digestive, respiratory, and circulatory systems in metazoans. So, for example, oxygen is obtained by simple diffusion in unicellular organisms but taken up by gills or lungs and transported through vascular systems in large metazoans.

A first step in understanding these transitions is to identify the fundamental energetic constraints on metabolic rate that change as a function of body size and across these three groups (Fig. 3). Inspired by the data, we propose the following hypotheses wherein different processes govern the scaling of metabolic rate in each group, and each evolutionary transition produced innovations in metabolic design that allowed further increases in body size and complexity:

1) Prokaryotes: We hypothesize that the very rapid increase in metabolic rate with increasing cell size is made possible by an increase in the number of genes. If cell size limits the number of genes and/or quantity of DNA, then larger cells can have larger genomes. In prokaryotes, larger genomes have more coding genes, which produce a larger number of different enzymes and result in larger, more complicated biochemical networks. These networks can confer increased metabolic power because cells can utilize a greater diversity of substrates as energy sources or use a given substrate more completely, thereby producing more ATP molecules per unit substrate and per unit time. For example, the quantity of energy resulting from the oxidation of a given substrate

during respiration is dependent on multiple factors, such as the type of terminal electron acceptor, the type of cytochrome system used to pump protons, and the efficiency of the ATP synthase. Within prokaryotes, the number of ATP molecules produced per oxygen molecule reduced by the respiratory chain has been observed to vary at least 3-fold (25). *Escherichia coli*, a prokaryote of average cell and genome size, can yield up to 1 or 2 ATP molecules per $\text{NADH} + \text{H}^+$, whereas yeast, which have larger, more complex metabolic networks presumably more comparable to the networks of the most complex prokaryotes, produce 3 ATP molecules per $\text{NADH} + \text{H}^+$ (26). Furthermore, different pathways of glycolysis produce differing amounts of ATP: some prokaryotes use the less efficient Entner-Doudoroff pathway, which produces 1 ATP per glucose molecule and is thought to be more primitive, whereas other prokaryotes and all eukaryotes use the Embden-Meyerhof pathway, which produces 2 ATP molecules per glucose molecule (26,27).

The link between cell size and metabolic network complexity is supported by three findings. First, genome size exhibits the predicted positive scaling with cell size. Fig. 4 shows that both number of genes and total genome size scale with cell size as $M^{0.35}$. The parallel scaling confirms that increasing genome size is due to increasing numbers of protein-coding genes (28). Second, limited data show a positive scaling relationship between the total number of metabolic reactions and genome size in prokaryotes ($R^2 = 0.83$, $y = 12.5x^{0.62}$, for the five taxa in Price et al. (29)). And third, the proportion of metabolism-related genes increases with genome size in prokaryotes (30).

These findings provide support for the idea that the superlinear scaling of metabolic rates in prokaryotes derives from the increase in genome size with body size.

Larger prokaryotes can attain increases in metabolic power by increasing genome size only up until the point when they have a relatively complete metabolic network. Since the respiratory complexes of enzymes and protein pumps used in ATP synthesis are located in the cell membranes of prokaryotes, the cell surface area must eventually limit metabolic rate, causing the metabolic scaling to decrease from superlinear to sublinear. This sets the stage for the evolutionary transition to protists, which because of their linear scaling, will tend to be more powerful and to outcompete similar-sized prokaryotes (31).

So a corollary prediction is that the few giant heterotrophic prokaryotes that overlap broadly in size with protists should occupy specialized ecological niches where they can have comparable metabolic rates and not be competitively excluded. Data for two giant prokaryotes support this prediction (17). *Thioploca araucae* has a mass of 2×10^{-8} g and a metabolic rate of 5.9×10^{-10} W, which is very close to the expected metabolic rate for a protist of that size at 3.4×10^{-10} W. Likewise, *Thiovulum majus* has a mass of 3×10^{-9} g and a metabolic rate of 9.9×10^{-11} W, also close to the expected rate at 5.4×10^{-11} W.

2) Protists: We hypothesize that the approximately linear scaling of metabolic rate in protists reflects a linear increase in the membrane bound sites of ATP synthesis located in organelles. The ancestral eukaryotes were able to overcome the constraints of limited ATP synthesizing sites on the cell surface by ingesting the symbiotic prokaryotes that evolved into mitochondria (32). This innovation allowed the host cell to overcome the constraint of limited cell surface area by containing many mitochondria, and hence a

much larger area of membrane with the requisite respiratory complexes composed of enzymes and proton pumps. This hypothesis predicts that the number or total volume of mitochondria scales linearly with cell mass, similar to the scaling of organs in metazoans. Support for this hypothesis comes from the linear relationship between mitochondrial volume and cell volume in the alga *Polytoma papillatum* (33) and yeast *Saccharomyces cerevisiae* (34), and the linear relationship between metabolic rates and the volume of mitochondria in cells of metazoans (35,36). Such linear scaling cannot be maintained indefinitely, however. As cell volume and number of mitochondria increase, capacity to supply resources to the respiratory complexes eventually becomes limiting, both because cell surface area limits the diffusion of some resources into the cell or the number of active sites for uptake of other resources from the environment, or because distances within the cell limit the delivery or diffusion of the resources to the mitochondria. The consequence is a transition from linear to sublinear scaling. The next evolutionary transition occurs at the size where the smallest metazoans begin to be more powerful and competitively superior to similar-sized protists.

3) Metazoans: We hypothesize that the scaling in the smallest metazoans will initially be near linear, as observed empirically in some animals and plants (15,37), because these smallest metazoans are composed of relatively few cells and minimal vascular or skeletal systems and are smaller than the size at which the scaling transitions to sublinear. However, this switch is difficult to see in our data. As body size increases, transport distances within organisms and exchanges of resources across surface areas increasingly come into play, and differentiated vascular systems evolved to collect and distribute

resources. Models of resource distribution through vascular networks show the impossibility of maintaining linear scaling of metabolic rate as body size increases in metazoans, and several different models independently suggest that the scaling converges to the $\alpha \approx 0.75$ of Kleiber's law (20,38).

The evolutionary transitions from prokaryotes to unicellular eukaryotes and to metazoans entailed structural and functional innovations that overcame constraints on their precursors, but imposed new constraints that governed the scaling of metabolic rate and limited maximum size in the newly evolved group. These transitions and the key innovations and constraints are diagrammed in Figure 3. Changes in the scaling of biological energetics over the 16 orders of magnitude in body size reflect the fundamental dependence of metabolic rate on: i) the number of membrane-bound respiratory complexes where proton pumping and ATP synthesis occur; and ii) geometric constraints on transport distances and surface exchanges that affect rates of resource supply. Because metabolism fuels biomass production for growth and reproduction, differences across the transitions in scaling of metabolism are reflected in differences in population growth rate and production efficiency. There is a need for additional research that examines each component of our mechanistic hypotheses. Of particular interest are the exceptional taxa that extend well into the size range of other groups, suggesting that they have evaded the typical constraints or occupied specialized ecological niches.

Our data and analyses clearly show that the sublinear metabolic scaling and quarter-power scaling relations documented for large, multicellular animals and plants, with the $\alpha \approx 0.75$ for metabolic rate and the $\alpha \approx -0.25$ for r_{max} , do not extend to the

smallest organisms. The evolution from prokaryotes to unicellular eukaryotes and then to multicellular organisms entailed major transitions in body size, metabolic processes, genomic organization, and overall complexity. Changes in allometric scaling relations across these transitions can help identify the most fundamental features of biological energetics that shaped the early evolution of life.

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Figure Legends

Figure 1. Relationship between whole organism metabolic rate and body mass for heterotrophic prokaryotes, protists, and metazoans plotted on logarithmic axes. Fits are shown with slopes corrected to RMA $\pm$ SE, active metabolic rate data in black and endogenous rate data in blue. Differences in slopes among all groups are significant for both physiological states ($p \leq 0.05$).

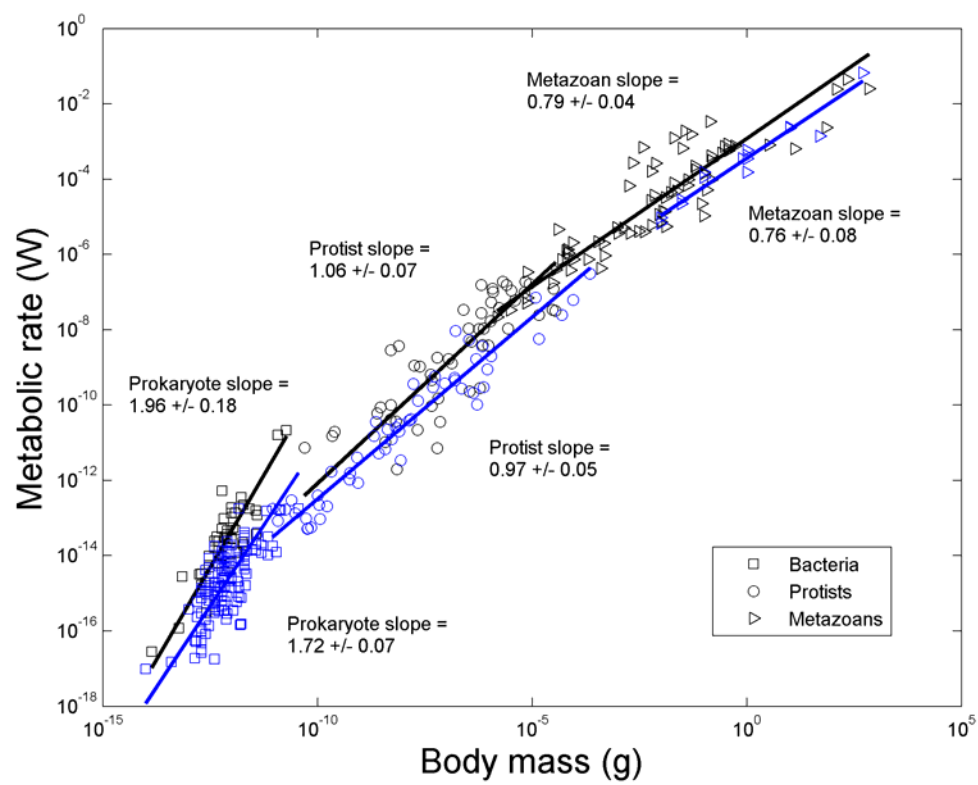
Figure 2. **(A)** Scaling of r_{max} (maximum rate of population growth; unfilled symbols) and mass-specific metabolic rate (B_{ms} , filled symbols) with body mass for heterotrophic prokaryotes, protists, and metazoans plotted on logarithmic axes. For r_{max} , the RMA-corrected slope is 0.73 for prokaryotes, -0.26 for protists, and -0.23 for metazoans. The plots for mass-specific metabolic rate are approximately parallel to those for r_{max} , consistent with the hypothesis that metabolic rate fuels biomass production. The apparent discrepancy in RMA (but not OLS) slopes in protists results from variability in protist metabolic rates. **(B)**. Efficiency of biomass production varies with body size across the three groups. Mean mass synthesized per unit of energy expended decreases over ten-fold (see text for means). Open symbols are those where r_{max} was known for a species but mass-specific metabolic rate was estimated, and closed symbols are for species where both r_{max} and mass-specific metabolic rate were known.

Figure 3. Schematic representation of our hypotheses explaining the metabolic scaling in prokaryotes, protists, and metazoans. Scaling within each group reflects constraints on

metabolic power due to the number of respiratory complexes, but geometric constraints on exchange surfaces and transport distances ultimately limit capacity to supply substrates to these respiratory complexes. Superlinear scaling in prokaryotes (solid blue line) reflects the increase in number of genes and metabolic enzymes with increasing cell size, until a new constraint (fading blue line) due to cell surface area, where the enzyme complexes and proton pumps are localized, becomes limiting, imposing sublinear scaling. Protists overcome this constraint by incorporating respiratory complexes into mitochondria. Larger protists can accommodate more organelles, resulting in metabolic rate scaling linearly with volume of mitochondria and cell mass (solid red line), until a new geometric constraint of surface exchange or transport distance limits rate of resource supply to the mitochondria, imposing sublinear metabolic scaling (fading red line). Metazoans face similar constraints, but can have greater metabolic power because they have delivery systems that keep cells supplied in larger aggregations. Vascular networks evolved to supply resources, but geometric constraints impose the approximately $\frac{3}{4}$ -power sublinear scaling of Kleiber's rule (green line).

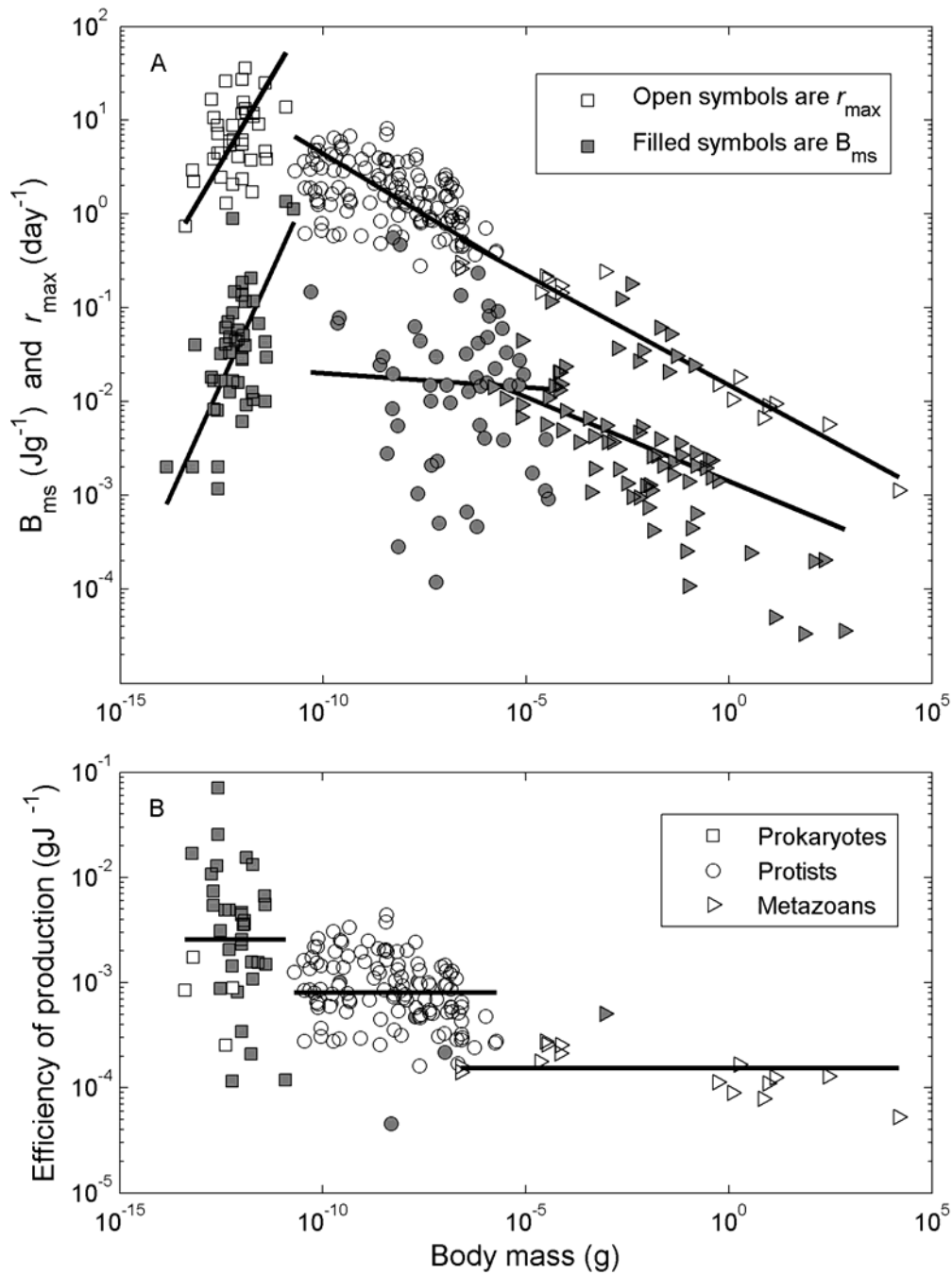
Figure 4. Scaling of genome size with cell size in prokaryotes. Total number of nucleotides (above) and number of different genes (below) scale with identical slopes of 0.35, consistent with our hypothesis that scaling of metabolic power in prokaryotes reflects the number of genes and the complexity of the biochemical network.

396 Figure 1.

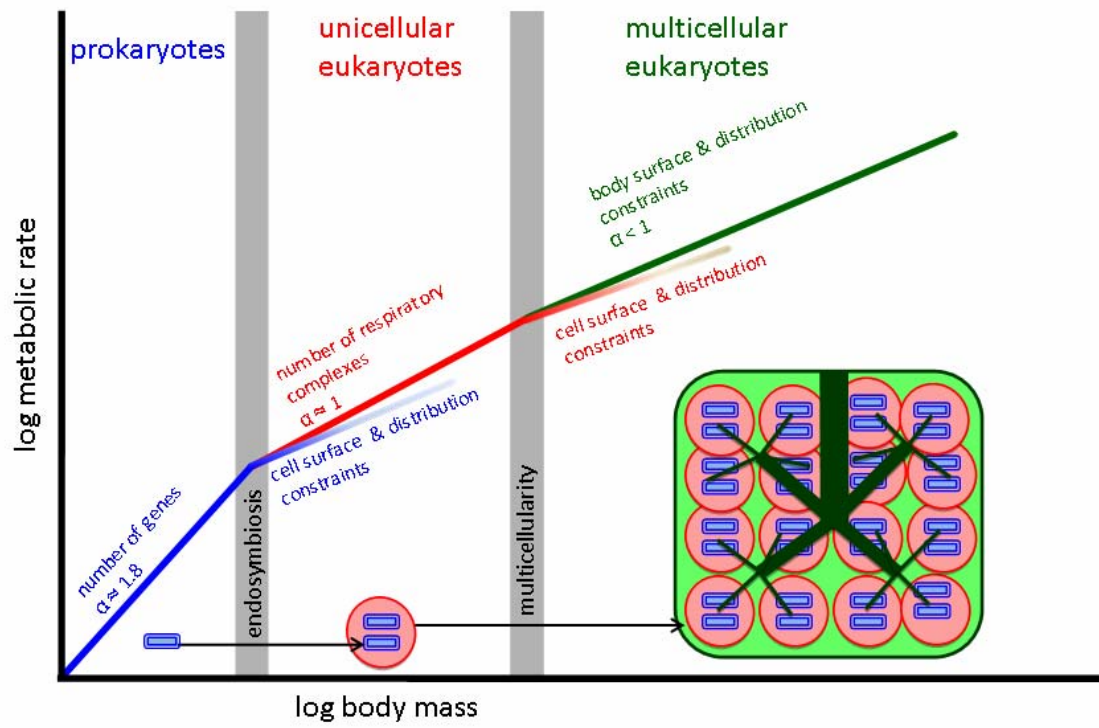


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398 Figure 2.

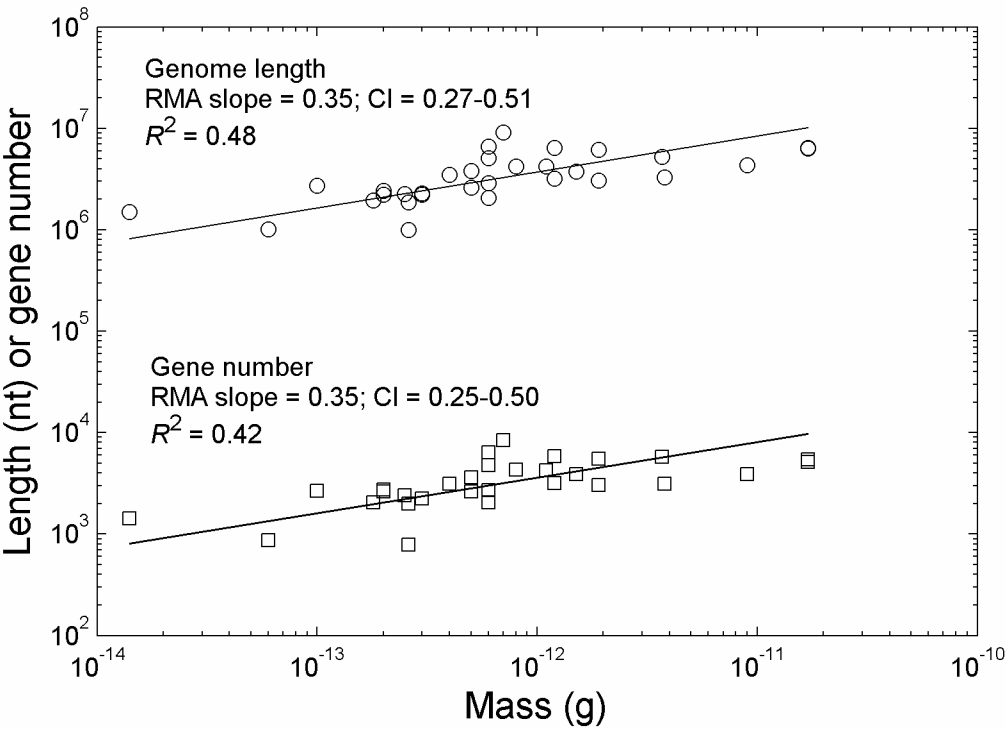


400 Figure 3.



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402 Figure 4.



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Supplementary Online Materials I. Methods

We combined metabolic rate data from several sources, and all data used in these analysis are available in Supplementary Online Materials II and III. Physiological state has a strong effect on metabolic rates and may influence the observed scaling of metabolic rate with mass (11)(39). We therefore separated data into active and endogenous rates. Active rates were defined as rates where individuals were measured in the presence of food or, if not, had only been washed free of their food just prior to measurement. Endogenous rates were defined as a wider range of resting, inactive, or starved states. We used the data sets compiled by Makareiva et al. (18) and (17) for active and endogenous rates, respectively, of prokaryotes, both of which are available as supplementary material attached to their original article. We only included prokaryotes species that are obligate heterotrophs (so excluded species capable of phototrophy, chemotrophy, and mixotrophy, and archaea). For active metabolic rates of eukaryotes, including protists and zooplankton, we surveyed the literature and developed new data sets. All values in these data sets were included only after consulting the original work and recording the original data and making determinations of physiological state of the individuals. For endogenous rates of protists, we used the data from Makareiva et al. (17), and for endogenous rates of small metazoans, we used the zooplankton data from Gillooly et al (40). Multiple values for a species were averaged to create a data set with one mass and one metabolic rate estimate per species. All original metabolic rate units were converted to W, and volumes and masses were converted to g. The active rate data set included 44, 52, and 71 species or

425 strains of prokaryotes, protists, and metazoans, respectively, and the endogenous rate data
426 set included 121, 52, and 15 species.

427 All organisms evaluated in this study are ectotherms, therefore temperature
428 strongly influences their metabolic rates. We used the Boltzmann factor with an
429 activation energy of 0.61 eV to correct all data to 20°C (40). This approach works well
430 because a single correction approach can be applied to all data, reducing the error
431 variance in the scaling estimates. We analyzed our metabolic rate data sets with
432 uncorrected data and still found superlinear scaling in prokaryotes, linear scaling in
433 protists, and sublinear scaling in metazoans, albeit with slightly shallower scaling
434 exponents.

435 The methods employed to measure body mass in these studies were diverse,
436 including actual weighing of animals, estimation of volume from external dimensions,
437 and by weighing large numbers of cells and dividing by an estimate of the number of
438 cells. In several cases for protists, body mass data were not available and were collected
439 from Fenchel and Finlay's estimates.

440 For all analyses we calculated scaling exponents with ANOVA on log-
441 transformed data. Differences in the slopes among groups were determined by comparing
442 models with group-by-slope interaction terms to models without these terms. The
443 presence of non-negligible error in the x-axis variable, however, indicates that the output
444 from an ordinary least squares (OLS) fitting procedure is likely to produce scaling slopes
445 that are artificially shallow. All previous studies on the scaling of unicells have used
446 uncorrected OLS, which is one reason that previous studies on the metabolic rate scaling

of protists reported sublinear slopes. As indicated above, there are several sources of error in the body masses reported in these data sets. Therefore, all OLS slopes were converted to the equivalent reduced major axis (RMA) slopes, because RMA apportions error equally to the independent and dependent variables. We corrected all slopes to the RMA equivalent by dividing by the correlation coefficient for mass and metabolic rate for each group.

We surveyed the literature to obtain $r_{\max}$ values for prokaryotes, and used data from Caron and Rose (41) for protists and Savage et al. (21) for metazoans. The data set included 37, 122, and 16 species or strains of prokaryotes, protists, and metazoans, respectively. We also collected genome size information for prokaryotes from the National Center for Biotechnology Information (NCBI) genome database (<http://www.ncbi.nlm.nih.gov/>). We checked all prokaryotes found in our active rate dataset (18), and found all species-level matches between that database and the NCBI database. For some species, multiple entries were present in the NCBI database, with varying genome sizes. In these cases, we always used the largest genome size. We extracted genome length and number of genes and paired these data with the body size and active metabolic rate data.

Supplementary Online Materials II. Metabolic rate data

IIa. Prokaryotes

Endogenous rates			Active rates		
Species	Mass (g)	Metabolic	Species	Mass (g)	Metabolic
		rate (W)			rate (W)
Francisella tularensis	1.00E-14	9.86E-18	Mycoplasma bovis	1.40E-14	2.80E-17
Acholeplasma laidlawii	4.00E-14	1.49E-17	Mycoplasma capricolum	6.00E-14	1.20E-16
Nocardia farcinica	1.00E-13	3.74E-16	Leptospira sp.	7.00E-14	2.81E-15
Haemophilus influenzae	1.40E-13	1.86E-17	Streptococcus pyogenes	1.80E-13	3.24E-15
Vibrio sp.	1.40E-13	5.66E-17	Lactococcus lactis	2.00E-13	3.34E-15
Methylophilus					
methylophilus	1.50E-13	5.14E-17	Neisseria gonorrhoeae	2.00E-13	1.65E-15
Paracoccus denitrificans	1.60E-13	5.99E-16	Streptococcus pneumoniae	2.50E-13	2.00E-15
Pseudomonas oleovorans	1.60E-13	2.38E-16	Mycoplasma gallisepticum	2.60E-13	5.20E-16
Achromobacter ruhlandii	2.00E-13	1.35E-15	Streptococcus thermophilus	2.60E-13	3.04E-16
Arthrobacter sp.	2.00E-13	1.00E-16	Neisseria meningitidis	3.00E-13	4.96E-15
Enterococcus faecalis	2.00E-13	2.71E-17	Streptococcus agalactiae	3.00E-13	9.75E-15
Lactococcus lactis	2.00E-13	5.41E-17	Aerobacter aerogenes	4.00E-13	2.45E-14
Mycobacterium tuberculosis	2.00E-13	1.17E-16	Enterococcus cecorum	4.00E-13	1.62E-14
Neisseria elongata	2.00E-13	7.49E-16	Vibrio sp. DW1	4.50E-13	3.19E-14
Neisseria flava	2.00E-13	1.08E-15	Bdellovibrio bacteriovorus	5.00E-13	6.32E-15
Neisseria gonorrhoeae	2.00E-13	3.20E-17	Corynebacterium sp.	5.00E-13	2.41E-14
Neisseria mucosa	2.00E-13	1.35E-15	Staphylococcus epidermidis	5.00E-13	1.67E-14
Neisseria sicca	2.00E-13	1.17E-15	Pseudomonas aeruginosa	6.00E-13	5.34E-13
Streptococcus pyogenes	2.00E-13	2.98E-16	Rhizobium leguminosarum	6.00E-13	1.00E-14
Nitrobacter winogradskyi	2.40E-13	1.08E-15	Zymomonas mobilis	6.00E-13	5.25E-14

Thiobacillus ferrooxidans	2.50E-13	8.34E-17	Burkholderia sp. JT1500	6.60E-13	9.83E-14
Thiobacillus thiooxidans	2.50E-13	1.98E-16	Bradyrhizobium sp.	7.00E-13	3.09E-14
Xanthomonas axonopodis	2.50E-13	4.17E-15	Bacillus licheniformis	8.00E-13	4.65E-14
Staphylococcus aureus	2.70E-13	1.15E-15	Enterococcus sp. RfL6	8.00E-13	3.72E-14
Acidovorax facilis	3.00E-13	9.47E-16	Lactococcus sp. TmLO5	8.00E-13	1.26E-14
Bdellovibrio bacteriovorus	3.00E-13	3.38E-15	Alteromonas haloplanktis	1.00E-12	1.87E-13
Brucella melitensis	3.00E-13	6.49E-16	Azospirillum brasilense	1.00E-12	2.80E-14
Enterobacter aerogenes	3.00E-13	1.35E-15	Bacillus macerans	1.00E-12	6.13E-15
Flavobacterium capsulatum	3.00E-13	8.53E-15	Pseudomonas natrigiens	1.00E-12	1.35E-13
Legionella pneumophila	3.00E-13	1.60E-15	Streptococcus faecalis	1.00E-12	2.92E-14
Streptococcus agalactiae	3.00E-13	1.36E-16	Bacillus subtilis	1.10E-12	5.57E-14
Enterococcus cecorum	4.00E-13	6.86E-16	Escherichia coli	1.20E-12	1.39E-13
Haemophilus parainfluenzae	4.00E-13	3.73E-16	Pseudomonas fluorescens	1.20E-12	4.73E-14
Halomonas halodenitrificans	4.00E-13	1.79E-14	Lactobacillus bulgaricus	1.30E-12	1.19E-14
			Pseudomonas		
Klebsiella pneumoniae	4.00E-13	5.95E-16	perfectomarinus	1.70E-12	3.51E-13
Mycobacterium phlei	4.00E-13	1.49E-15	Desulfovibrio propionicus	1.80E-12	2.27E-14
Proteus morganii	4.00E-13	9.02E-16	Lactobacillus casei	1.90E-12	1.97E-14
Proteus vulgaris	4.00E-13	5.33E-16	Pseudomonas putida	1.90E-12	2.23E-13
Sallinivibrio costicola	4.00E-13	1.87E-15	Vibrio anguillarum	2.60E-12	1.77E-13
Serratia marcescens	4.00E-13	4.26E-16	Bacillus cereus	3.70E-12	1.60E-13
Streptococcus pneumoniae	4.00E-13	3.07E-16	Lactobacillus plantarum	3.80E-12	3.77E-14
Taylorella equigenitalis	4.00E-13	1.80E-17	Azospirillum lipoferum	4.00E-12	1.19E-13
Thiobacillus intermedius	4.00E-13	1.98E-15	Azotobacter chroococcum	1.20E-11	1.62E-11
Achromobacter xerosis	5.00E-13	5.19E-15	Azotobacter agilis	1.90E-11	2.14E-11
Arthrobacter globiformis	5.00E-13	1.87E-15			

<i>Azorhizobium caulinodans</i>	5.00E-13	8.57E-15
<i>Azotobacter vinelandii</i>	5.00E-13	3.38E-16
<i>Pseudomonas aeruginosa</i>	5.00E-13	1.51E-15
<i>Staphylococcus epidermidis</i>	5.00E-13	6.09E-15
<i>Yersinia pestis</i>	5.50E-13	1.39E-15
<i>Achromobacter</i> sp.	6.00E-13	2.27E-15
<i>Achromobacter viscosus</i>	6.00E-13	9.47E-15
<i>Aminobacter lissarensis</i>	6.00E-13	6.40E-16
<i>Nitrosomonas europaea</i>	6.00E-13	4.40E-15
<i>Rhizobium leguminosarum</i>	6.00E-13	2.57E-15
Unidentified bacterium	6.00E-13	2.98E-15
<i>Vibrio alginolyticus</i>	6.00E-13	8.93E-16
<i>Vibrio fischeri</i>	6.00E-13	1.32E-14
<i>Vibrio metschnikovii</i>	6.00E-13	8.93E-16
<i>Vibrio parahaemolyticus</i>	6.00E-13	4.60E-16
<i>Salmonella typhimurium</i>	6.60E-13	2.64E-15
<i>Bacillus pumilus</i>	7.00E-13	1.58E-15
<i>Bacillus stearothermophilus</i>	7.00E-13	6.11E-16
<i>Bradyrhizobium japonicum</i>	7.00E-13	3.41E-16
<i>Burkholderia</i> sp.	7.00E-13	6.63E-15
<i>Escherichia coli</i>	7.00E-13	5.97E-15
<i>Rhizobium japonicum</i>	7.00E-13	8.77E-15
<i>Rhizobium meliloti</i>	7.00E-13	3.79E-15
<i>Acetobacter aceti</i>	7.50E-13	2.03E-16
<i>Alcaligenes eutrophus</i>	8.00E-13	2.38E-14
<i>Bacillus popilliae</i>	8.00E-13	2.89E-16

<i>Delftia acidovorans</i>	8.00E-13	4.69E-15
<i>Enterococcus</i> sp.	8.00E-13	1.19E-14
<i>Lactococcus</i> sp.	8.00E-13	8.30E-16
<i>Rhodobacter sphaeroides</i>	8.00E-13	7.16E-15
<i>Aquaspirillum itersonii</i>	9.00E-13	4.06E-15
<i>Bacillus firmus</i>	9.00E-13	5.28E-15
<i>Enterobacter cloacae</i>	9.00E-13	1.26E-14
<i>Azospirillum brasiliense</i>	1.00E-12	7.19E-15
<i>Methylobacterium</i>		
<i>extorquens</i>	1.00E-12	2.71E-15
<i>Methylosinus trichosporium</i>	1.00E-12	1.89E-14
<i>Myxococcus xanthus</i>	1.00E-12	2.08E-15
<i>Picrophilus oshimae</i>	1.00E-12	1.38E-15
<i>Pseudomonas fluorescens</i>	1.00E-12	2.32E-15
<i>Thiocapsa roseopersicina</i>	1.00E-12	2.53E-15
<i>Micrococcus luteus</i>	1.10E-12	3.52E-16
<i>Branhamella catarrhalis</i>	1.30E-12	5.89E-16
<i>Bacillus subtilis</i>	1.40E-12	2.08E-15
<i>Agrobacterium tumefaciens</i>	1.50E-12	1.35E-14
<i>Arthrobacter</i> sp.	1.50E-12	8.12E-16
<i>Beneckea natriegens</i>	1.50E-12	1.79E-13
<i>Chromatium vinosum</i>	1.50E-12	1.49E-15
<i>Desulfovibrio salexigens</i>	1.50E-12	8.80E-15
<i>Lactobacillus brevis</i>	1.60E-12	1.44E-16
<i>Arthrobacter</i>		
<i>crystallopoietes</i>	1.70E-12	1.53E-16

<i>Pseudomonas putida</i>	1.70E-12	7.67E-16
<i>Acinetobacter baumannii</i>	2.00E-12	1.08E-14
<i>Acinetobacter calcoaceticus</i>	2.00E-12	6.04E-15
<i>Acinetobacter johnsonii</i>	2.00E-12	4.15E-14
<i>Acinetobacter</i> sp.	2.00E-12	4.40E-15
<i>Acinetobacter</i> sp.	2.00E-12	5.41E-15
<i>Acinetobacter</i> sp.	2.00E-12	7.22E-15
<i>Cellvibrio gilvus</i>	2.00E-12	3.34E-14
<i>Phaeospirillum fulvum</i>	2.00E-12	2.95E-14
<i>Nocardia corallina</i>	2.10E-12	1.61E-15
<i>Pediococcus acidilactici</i>	2.20E-12	3.28E-15
<i>Moraxella osloensis</i>	2.30E-12	1.35E-14
<i>Methylobacterium</i> sp.	3.00E-12	1.49E-14
<i>Bacillus cereus</i>	3.70E-12	1.38E-14
<i>Sporosarcina ureae</i>	3.80E-12	2.06E-14
<i>Azospirillum lipoferum</i>	4.00E-12	4.16E-14
<i>Amoebobacter roseus</i>	5.00E-12	1.22E-14
<i>Amoebobacter pendens</i>	5.00E-12	1.92E-14
<i>Sphaerotilus natans</i>	6.50E-12	1.54E-13
<i>Bacillus megaterium</i>	7.00E-12	1.04E-14
<i>Pseudomonas formicans</i>	7.00E-12	3.16E-14
<i>Rhodospirillum rubrum</i>	9.00E-12	1.80E-14
<i>Thiocystis violacea</i>	1.10E-11	1.24E-14
<i>Azomonas agilis?</i>	1.30E-11	1.68E-13
<i>Azotobacter chroococcum</i>	1.40E-11	1.58E-13
<i>Amoebobacter purpureus</i>	3.60E-11	1.79E-13

Source for endogenous prokaryote rates

1. A. M. Makarieva et al., *Proc Natl Acad Sci U S A.* **105**, 16994–16999 (2008).

Source for active prokaryote rates

1. A. Makarieva, V. Gorshkov, B. Li, *Proceedings: Biological Sciences* **272**, 2219-2224 (2005).

Iib. Protists

Endogenous rates			Active rates		
Species	Mass (g)	Metabolic	Species	Mass (g)	Metabolic
		rate (W)			rate (W)
Acanthamoeba castellani	6.70E-09	2.55E-11	Acanthamoeba castellanii	2.95E-09	8.85E-11
Acanthamoeba sp.	4.32E-09	2.29E-11	Acanthamoeba palestinensis	6.76E-09	3.75E-11
Actinosphaerium eichhornii	1.46E-05	5.82E-09	Acanthamoeba rhyodes	5.00E-09	4.18E-11
Amoeba proteus	9.00E-07	9.00E-10	Ammonia sp.	6.54E-06	9.86E-08
Astasia klebsii	3.80E-09	6.84E-12	Amoeba proteus	9.55E-07	3.90E-09
Astasia longa	1.35E-08	3.92E-11	Amoeba sp.	2.12E-08	2.22E-11
Bresslaua insidiatrix	1.70E-08	3.69E-10	Astasia klebsii	3.75E-09	1.05E-11
Chilomonas paramecium	2.26E-09	3.62E-11	Astasia longa	2.47E-09	6.07E-11
Coleps hirtus	9.10E-08	3.82E-10	Blepharisma americanum	2.50E-07	3.38E-08
Colpidium campylum	5.44E-08	5.88E-10	Bolivina pacifica	1.09E-06	1.73E-08
Crithida fasciculata	2.08E-09	1.52E-11	Bolivina spissa	4.00E-06	5.96E-08
Crithidia oncopelti	3.00E-11	1.35E-13	Bresslaua insidiatrix	2.40E-08	1.07E-09
Dictyostelium discoideum	8.40E-10	4.12E-12	Bulimina subornata	1.71E-06	3.84E-08
Eimeria acervulina	2.64E-09	5.28E-12	Buliminella sp.	6.60E-07	1.55E-07
Eimeria stiedae	7.98E-09	2.00E-11	Chaos carolinense	3.02E-05	3.37E-08
Eimeria tenella	5.44E-09	1.25E-11	Chaos carolinensis	3.50E-05	3.18E-08
Endotrypanum schaudinni	1.14E-10	9.58E-14	Chilomonas paramecium	5.14E-09	1.02E-10
Entamoeba histolytica	8.60E-09	3.44E-12	Chilostomella ovoidea	1.44E-05	2.50E-08
Frontonia leucas	7.45E-07	2.98E-10	Cibicidoides woellerstorfi	6.89E-06	1.90E-07
Leishmania brasiliensis	9.00E-12	1.80E-13	Colpidium campylum	4.30E-08	6.44E-10
Leishmania donovani	1.80E-11	1.67E-13	Colpoda cucullus	4.50E-08	4.58E-10
Leishmania enrietti	1.20E-11	8.40E-14	Corythion dubium	6.54E-08	1.52E-10

Mayorella palestinensis	8.30E-09	3.07E-11	Didinium nasutum	7.35E-07	1.08E-08
Noctiluca miliaris	2.24E-04	3.14E-07	Elphidium sp.	1.19E-06	1.24E-07
Paramecium aurelia	1.60E-07	4.64E-10	Entamoeba hystolitica	8.60E-06	1.66E-07
Paramecium calkinsi	1.57E-07	5.34E-10	Favella ehrenbergii	6.60E-07	2.78E-08
Paramecium caudatum	5.26E-07	1.05E-10	Favella taraikaensis	3.30E-07	1.07E-08
Paramecium					
multimicronucleatum	6.85E-07	3.63E-09	Frontonia leucas	6.14E-07	2.86E-10
Pelomyxa carolinensis	4.97E-05	2.49E-08	Globobulimina affinis	3.11E-05	1.23E-07
Pelomyxa palustris	9.00E-05	6.30E-08	Ochromonas sp.	2.50E-10	1.96E-11
Plasmodium cathemerium	5.40E-11	9.72E-14	Paramecium aurelia	1.12E-07	1.68E-09
Plasmodium gallinaceum	7.10E-11	5.68E-14	Paramecium calkinsi	1.35E-07	1.31E-09
Plasmodium knowlesi	5.90E-11	5.31E-14	Paramecium caudatum	5.81E-07	1.06E-08
Pleuromonas jaculans	2.50E-11	3.00E-13	Paraphysomonas imperforata	2.22E-10	1.52E-11
Podophrya fixa	1.49E-08	4.32E-11	Placus sp. 1	3.82E-07	4.89E-09
Schizotrypanum					
verpertilionis	1.08E-10	2.27E-13	Placus sp. 2	7.12E-07	4.00E-09
Spirostoma minus	5.00E-07	1.70E-09	Pleuromonas jaculans	5.00E-11	7.46E-12
Spirostomum ambiguum	1.20E-05	7.08E-08	Podophrya fixa	4.55E-08	9.48E-11
Spirostomum intermedium	2.29E-07	2.75E-10	Polychaos fasciculatum	7.13E-08	3.59E-11
Spirostomum teres	4.23E-07	2.12E-10	Qunqueoloculina sp.	2.06E-06	1.88E-07
Stentor coeruleus	1.10E-06	1.98E-09	Reophax sp.	2.56E-06	1.54E-07
Tetrahymena pyriformis	2.20E-08	1.30E-10	Saccamoeba limax	5.15E-09	2.89E-09
Tetrahymena pyriformis	4.90E-08	2.99E-10	Spirostomum ambiguum	2.74E-06	1.07E-08
Tracheloraphis sp.	3.40E-07	5.44E-09	Spirostomum teres	3.49E-07	2.31E-10
Trichomonas foetus	5.80E-10	1.57E-12	Stainforthia apertura	1.11E-06	5.38E-08
Trichomonas batrachorum	5.60E-10	1.06E-12	Tetrahymena pyriformis	1.79E-08	1.13E-09

Trichomonas nasai	2.10E-10	1.70E-12	Textularia kattedagensis	1.24E-06	1.02E-07
Trichomonas vaginalis	8.70E-10	8.70E-13	Tintinnopsis acuminata	7.07E-09	1.98E-12
Trypanosoma cruzi	1.69E-10	2.03E-13	Tintinnopsis vasculum	6.08E-08	7.17E-12
Trypanosoma lewisi	5.80E-11	5.22E-14	Trichosphaerium sieboldi	6.10E-08	1.85E-09
Urostyla grandis	1.66E-07	9.46E-09	Uvigerina akitaensis	3.30E-06	1.10E-07
Trichosphaerium sieboldi	1.00E-10	4.00E-13	Vanella sp.	7.93E-09	3.73E-09

Source for endogenous protist rates

1. A. M. Makarieva et al., *Proc Natl Acad Sci U S A.* **105**, 16994–16999 (2008).

Sources for active protist rates

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IIc. Metazoans

Endogenous rates			Active rates		
Species	Mass (g)	Metabolic	Species	Mass (g)	Metabolic
		rate (W)			rate (W)
Asterias rubens	1.00E+01	2.23E-03	Acartia clausi	4.01E-05	4.63E-06
Parastichopus japonicus	5.00E+01	1.40E-03	Asellus aquaticus	1.68E-02	4.42E-05
Homarus americanus	5.00E+02	6.70E-02	Bathycalanus sp.	1.38E-01	3.34E-03
Pachygrapsus crassipes	1.00E+01	2.40E-03	Bosmina longirostris	7.69E-06	7.00E-08
Uca pugilator	7.84E-01	3.52E-04	Brachionus calyciflorus	1.54E-06	2.16E-08
Erpobdella octoculata	3.00E-02	2.23E-05	Brachionus plicatilis	3.08E-06	3.31E-08
			Brachyuran larvae		
Glossiphonia complanata	3.00E-02	2.76E-05	(megalops)	9.00E-03	1.14E-05
			Branchinella		
Arenicola marina	1.00E+00	3.46E-04	kugenumaensis	9.96E-02	1.06E-05
Clymenella torquata	1.00E-01	1.10E-04	Calanoides carinatus	1.00E-03	3.70E-06
Mercenaria mercenaria	1.00E+00	5.92E-04	Calanus finmarchicus	1.78E-03	6.53E-05
Mytilus edulis	1.00E+00	4.27E-04	Calanus gracilis	4.15E-03	3.97E-06
Crepidula fornicata	1.00E+00	1.52E-04	Calanus hyperboreus	1.99E-02	7.82E-05
Helix pomatia	1.00E-01	1.51E-04	Calanus pacificus (F)	1.38E-03	5.03E-06
Crenobia alpina	1.00E-02	6.70E-06	Calanus pacificus (II)	7.69E-05	3.75E-07
Polycelis felina	1.00E-02	9.49E-06	Calanus pacificus (IV)	2.00E-04	7.33E-07
			Calanus pacificus (N1)	7.69E-06	3.41E-07
			Calanus pacificus (V)	3.38E-04	2.21E-06
			Calanus sp.	1.22E-02	1.37E-05
			Candacia spp.	1.92E-03	3.61E-06
			Cavolinia inflexa	1.00E-02	7.34E-06

Centropages hamatus	7.01E-05	1.07E-06
Centropages typicus	6.63E-05	1.34E-06
Daphnia ambigua	3.54E-05	3.88E-07
Daphnia carinata	4.82E-04	9.22E-07
Daphnia galeata	9.23E-05	7.30E-07
Daphnia magna	9.39E-04	5.17E-06
Daphnia pulex	4.62E-04	1.95E-06
Diacria trispinosa	1.17E-01	5.16E-05
Epilabidocera amphitrites	3.85E-03	6.92E-04
Euchaeta spp.	6.23E-03	5.90E-06
Euchirella rostrata	5.94E-03	1.59E-04
Euclio pyramidata	2.31E-02	4.78E-05
Euphausia pacifica	1.52E-01	3.13E-04
Eurytemora herdmanni	4.94E-05	7.24E-07
Euthemisto compressa	3.64E-02	1.90E-03
Gammarus fossarum	4.19E-02	6.90E-05
Gammarus lacustris	1.51E-01	9.56E-05
Gammarus pulex	6.58E-02	1.75E-04
Gammarus roeseli	9.92E-02	1.38E-04
Homarus americanus	2.18E+02	4.39E-02
Hyas araneus	1.30E+01	6.43E-04
Hyperia galba	5.10E-02	1.55E-03
Jasus edwardsii	7.00E+02	2.48E-02
Libinia emarginata	1.23E+02	2.43E-02
Limacina helicoides	1.31E-02	5.48E-06
Meganyctiphanes norvegica	2.58E-01	4.97E-04

<i>Nematoscelis atlantica</i>	7.00E-03	3.74E-05
<i>Niphargus sphagnicolus</i>	5.90E-03	2.72E-05
<i>Oithona similis</i>	7.46E-06	5.07E-08
<i>Oncaea</i> sp.	3.08E-05	1.76E-07
<i>Pagrus major</i>	2.00E-02	1.23E-03
<i>Palaemon peringueyi</i>	1.46E-01	4.14E-04
<i>Paraeuchaeta norvegica</i>	3.15E-02	6.48E-04
<i>Parathemisto pacifica</i>	1.27E-02	3.27E-05
<i>Phronima sedentaria</i>	3.05E-01	7.33E-04
<i>Pleurobrachia bacheii</i>	8.77E-02	2.22E-05
<i>Pleuromamma gracilis</i>	3.08E-03	4.13E-06
<i>Pleuromamma robusta</i>	2.18E-03	2.71E-04
<i>Pleuroncodes planipes</i>	3.32E+00	8.04E-04
<i>Pseudocalanus minutus</i>	6.32E-05	8.33E-07
<i>Rhincalanus nasutus</i>	7.50E-03	2.59E-04
<i>Sagitta elegans</i>	1.10E-02	1.41E-05
<i>Sergestes</i> sp.	5.29E-01	7.48E-04
<i>Simocephalus vetulus</i>	3.96E-04	4.22E-07
<i>Solea senegalensis</i>	7.00E+01	2.31E-03
<i>Systellaspis debilis</i>	3.69E-01	5.72E-04
<i>Temora longicornis</i>	6.17E-05	1.25E-06
<i>Themisto</i> sp.	6.15E-02	2.21E-04
<i>Tomopteris septentrionalis</i>	4.32E-02	9.66E-05
<i>Tortanus discaudalus</i>	8.90E-05	2.08E-06
<i>Triops longicaudatus</i>	3.64E-01	8.50E-04

Source for endogenous metazoan rates

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Supplementary Online Materials III. $r_{\max}$ data

IVa. Prokaryotes

Species	Body mass (g)	$r_{\max}$ (day ⁻¹)
<i>Aerobacter aerogenes</i>	4.00E-13	26.03
<i>Alteromonas haloplanktis</i>	1.00E-12	5.56
<i>Azospirillum brasilense</i>	1.00E-12	6.18
<i>Azospirillum lipoferum</i>	4.00E-12	3.85
<i>Azotobacter chroococcum</i>	1.20E-11	13.88
<i>Bacillus cereus</i>	3.70E-12	25.02
<i>Bacillus licheniformis</i>	8.00E-13	4.09
<i>Bacillus macerans</i>	1.00E-12	2.35
<i>Bacillus subtilis</i>	1.10E-12	15.61
<i>Bdellovibrio bacteriovorus</i>	5.00E-13	5.36
<i>Corynebacterium glutamicum</i>	6.19E-13	6.25
<i>Desulfovibrio propionicus</i>	1.80E-12	1.71
<i>Escherichia coli</i>	1.20E-12	36.02
<i>Lactobacillus bulgaricus</i>	1.30E-12	12.17
<i>Lactobacillus casei</i>	1.90E-12	11.85
<i>Lactobacillus plantarum</i>	3.80E-12	4.69
<i>Lactococcus lactis</i>	2.00E-13	10.72
<i>Leptospira biflexa</i>	4.07E-13	1.29
<i>Mycoplasma capricolum</i>	6.00E-14	2.92

<i>Mycoplasma gallisepticum</i>	2.60E-13	4.43
<i>Mycoplasma pneumoniae</i>	4.00E-14	0.74
<i>Mycoplasma pulmonis</i> UAB CTIP	6.54E-14	2.22
<i>Neisseria gonorrhoeae</i>	2.00E-13	3.88
<i>Neisseria meningitidis</i>	3.00E-13	4.43
<i>Pseudomonas aeruginosa</i>	6.00E-13	8.86
<i>Pseudomonas fluorescens</i>	1.20E-12	13.15
<i>Pseudomonas natriegens</i>	1.00E-12	27.13
<i>Pseudomonas perfectomarinus</i>	1.70E-12	3.75
<i>Pseudomonas putida</i>	1.90E-12	10.93
<i>Rhizobium leguminosarum</i>	6.00E-13	2.07
<i>Staphylococcus epidermidis</i>	5.00E-13	5.96
<i>Streptococcus agalactiae</i>	3.00E-13	2.46
<i>Streptococcus faecalis</i>	1.00E-12	11.81
<i>Streptococcus pneumoniae</i>	2.50E-13	8.86
<i>Streptococcus pyogenes</i>	1.80E-13	16.68
<i>Streptococcus thermophilus</i>	2.60E-13	7.18
<i>Vibrio anguillarum</i>	2.60E-12	9.10

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IVb. Protists

Species	Body mass (g)	$r_{\max}$ (day⁻¹)
<i>Acanthamoeba polyphaga</i>	1.31E-09	4.86
<i>Actinomonas mirabilis</i>	7.50E-11	5.76
<i>Amoeba algonquinensis</i>	8.60E-09	0.57
<i>Amphidinium</i> sp.	2.60E-09	0.85
<i>Arcella vulgaris</i>	2.72E-07	0.47
<i>Aspidisca angulata</i>	3.80E-09	3.70
<i>Aspidisca</i> sp	7.10E-09	0.98
<i>Balanion comatum</i>	2.30E-09	4.00
<i>Balanion planctonicum</i>	2.26E-09	2.58
<i>Bodo designis</i>	5.40E-11	3.84
<i>Bodo saliens</i>	9.00E-11	0.80
<i>Bodo saltans</i>	7.40E-11	4.80
<i>Bursaridium difficile</i>	1.52E-07	0.85
<i>Caecitellus parvulus</i>	4.80E-11	1.90
<i>Cafeteria roenbergensis</i>	1.48E-10	1.27
<i>Chilodonella uncinata</i>	6.17E-09	1.92
<i>Clydonella rosenfieldi</i>	4.66E-10	1.34
<i>Cochliopodium minus</i>	3.25E-09	1.60
<i>Colpidium campylum</i>	2.15E-08	1.33
<i>Condyllostoma arenarium</i>	1.76E-06	0.39

<i>Condylostoma patulum</i>	5.48E-07	0.37
<i>Cyclidium glaucoma</i>	2.44E-09	2.37
<i>Cyclidium</i> sp.	8.87E-10	3.84
<i>Dactylamoeba</i> sp.	2.59E-09	0.49
<i>Diophrys scutum</i>	1.49E-07	0.93
<i>Entosiphon sulcatum</i>	1.08E-09	3.12
<i>Euplotes antarcticus</i>	2.60E-07	1.03
<i>Euplotes balteatus</i>	1.20E-08	3.63
<i>Euplotes crassus</i>	1.39E-07	1.96
<i>Euplotes eurystomus</i>	2.60E-07	0.45
<i>Euplotes focardii</i>	2.60E-07	0.93
<i>Euplotes harpa</i>	1.96E-07	0.46
<i>Euplotes minuta</i>	8.30E-08	2.31
<i>Euplotes</i> sp.	2.60E-07	0.51
<i>Euplotes trisulcatus</i>	9.97E-08	0.53
<i>Euplotes vannus</i>	1.90E-07	1.74
<i>Euplotes woodruffii</i>	2.60E-07	0.67
<i>Fabrea salina</i>	1.02E-06	0.71
<i>Favella azorica</i>	1.02E-07	2.40
<i>Favella ehrenbergii</i>	1.02E-07	0.79
<i>Favella</i> sp.	1.02E-07	1.55
<i>Favella taraikaensis</i>	1.02E-07	2.60

<i>Glaseria mira</i>	4.17E-10	3.65
<i>Gymnodinium</i> sp.	1.68E-10	1.64
<i>Gyrodinium dominans</i>	2.51E-08	1.67
<i>Gyrodinium fusiforme</i>	2.40E-08	0.84
<i>Halteria</i> sp.	1.06E-08	1.37
<i>Holosticha</i> sp	1.17E-07	0.85
<i>Hymenostome ciliate</i>	1.90E-08	4.25
<i>Jakoba libera</i>	3.50E-11	0.62
<i>Katodinium glaucum</i>	2.40E-08	0.28
<i>Keronopsis rubra</i>	7.11E-08	0.51
<i>Lacrymaria marina</i>	5.70E-09	0.65
<i>Litonotus lamella</i>	5.70E-09	1.41
<i>Lohmanniella spiralis</i>	1.50E-07	2.09
<i>Loxocephalus plagius</i>	3.31E-08	1.19
<i>Mayorella</i> sp.	5.58E-09	1.35
<i>Metachaos</i> sp.	1.15E-08	1.23
<i>Monas</i> sp.	3.40E-11	3.66
<i>Monosiga ovata</i>	6.20E-11	1.75
<i>Monosiga</i> sp	2.00E-11	2.88
<i>Ochromonas</i> sp.	1.56E-10	4.32
<i>Oxyrrhis marina</i>	1.40E-09	1.32
<i>Paraflabellula reniformis</i>	4.51E-10	1.40

<i>Paramecium aurelia</i>	2.33E-07	0.96
<i>Paramecium bursaria</i>	2.18E-07	0.82
<i>Paraphysomonas bandaiensis</i>	7.00E-11	1.42
<i>Paraphysomonas imperforata</i>	2.60E-10	6.00
<i>Paraphysomonas imperforata</i> (arctic)*	2.74E-10	5.20
<i>Paraphysomonas imperforata</i> (newfoundland)*	2.04E-10	4.15
<i>Paraphysomonas</i> sp.	1.80E-10	6.49
<i>Paraphysomonas vestita</i>	2.90E-10	5.04
<i>Paratetrahymena wassi</i>	2.60E-08	0.86
<i>Parauremonema acutum</i>	3.64E-09	3.09
<i>Pelagostrombidium fallax</i>	5.00E-08	0.86
<i>Pfiesteria piscicida</i>	2.50E-10	0.61
<i>Platyamoeba australis</i>	2.41E-10	1.94
<i>Platyamoeba</i> sp	3.53E-11	1.90
<i>Pseudobalanion planctonicum</i>	1.80E-09	2.13
<i>Pseudobodo tremulans</i>	9.00E-11	3.12
<i>Pteridomonas danica</i>	4.80E-11	3.02
<i>Rhizamoeba</i> sp.	1.80E-10	0.59
<i>Rimostrombidium caudatum</i>	4.20E-08	0.93
<i>Rimostrombidium veniliae</i>	4.20E-08	1.61
<i>Saccamoeba limax</i>	4.79E-09	2.20
<i>Scuticociliate</i>	3.70E-09	8.26

<i>Spirostomum teres</i>	2.15E-07	0.27
<i>Spumella</i> sp.	6.30E-11	4.80
<i>Stentor polymorphus</i>	1.85E-06	0.40
<i>Stephanoeca diplocostata</i>	7.50E-11	1.27
<i>Stereomyxa ramosa</i>	8.57E-10	0.59
<i>Strobilidium gyrans</i>	4.20E-08	0.87
<i>Strobilidium lacustris</i>	1.13E-07	1.56
<i>Strobilidium neptuni</i>	1.10E-07	2.09
<i>Strobilidium veniliae</i>	1.96E-08	1.01
<i>Strombidinopsis acuminatum</i>	1.40E-07	1.39
<i>Strombidinopsis</i> sp.	1.40E-07	0.95
<i>Strombidium capitatum</i>	6.41E-08	1.16
<i>Strombidium reticulatum</i>	4.00E-08	1.69
<i>Strombidium</i> sp.	2.50E-08	0.81
<i>Strombidium sulcatum</i>	7.01E-09	2.40
<i>Tetrahymena pyriformis</i>	8.59E-09	2.05
<i>Tetryhymena pyriformis</i>	1.93E-08	3.84
UnID chrysomonad	1.80E-10	2.94
UNID kinetoplastid	9.00E-11	0.67
Unidentified amoeba	7.20E-11	3.39
<i>Uronema elegans</i>	7.21E-09	2.66
<i>Uronema marina</i>	4.50E-10	6.80

<i>Uronema marinum</i>	3.59E-09	7.08
<i>Uronema nigricans</i>	3.60E-09	3.82
<i>Uronema</i> sp.	3.24E-09	3.80
<i>Urotricha castalia</i>	9.75E-09	1.28
<i>Urotricha farcta</i>	6.82E-09	3.59
<i>Urotricha furcata</i>	9.05E-09	1.69
<i>Vahlkampfia baltica</i>	3.78E-10	1.08
<i>Vahlkampfia damariscottae</i>	8.75E-11	1.88
<i>Vanella</i> sp.	7.90E-09	1.50
<i>Vannella caledonica</i>	4.02E-10	1.61
<i>Vannella</i> sp.	7.47E-11	1.37
<i>Vexillifera bacillipedes</i>	3.15E-10	2.89
<i>Vorticella microstoma</i>	2.15E-08	2.61
<i>Vorticella similis</i>	2.40E-08	2.26

Source for Protists

1. J. Rose, D. Caron, *Limnology and Oceanography* **52**, 886-895 (2007).

IVc. Metazoans

Species	Body mass (g)	$r_{\max}$ (day ⁻¹)
<i>Alburnus alburnus</i>	6.92E+06	0.01
<i>Ceriodaphnia dubia</i>	3.36E+01	0.20
<i>Chydorus sphaericus</i>	3.00E+01	0.22
<i>Daphnia magna</i>	8.85E+02	0.24
<i>Etheostoma flabellare</i>	1.23E+06	0.01
<i>Etheostoma spectabile</i>	5.38E+05	0.02
<i>Eurycercus longirostris</i>	7.14E+01	0.17
<i>Eurycercus vernalis</i>	7.00E+01	0.14
<i>Filinia pejleri</i>	2.50E-01	0.26
<i>Filinia terminalis</i>	2.50E-01	0.30
<i>Gadus morhua</i>	1.50E+10	0.00
<i>Gobio gobio</i>	9.23E+06	0.01
<i>Hippoglossoides platessoides</i>	2.77E+08	0.01
<i>Leuciscus leuciscus</i>	1.38E+07	0.01
<i>Pimephales promelas</i>	1.77E+06	0.02
<i>Pleuroxus denticulatis</i>	2.27E+01	0.15

Source for Metazoans

1. V. M. Savage, J. F. Gillooly, J. H. Brown, G. B. West, E. L. Charnov, *The American Naturalist* **163**, 429-441 (2004).